GoodNight: Nighttime Safety Prediction System for Women, Young Adults, and Urban Travelers

Github repo | Video Demo
Alyssa Velasquez (aav103), Jiya Patel (jtp158), and Roseanne Pham (np1029)

Project Definition

GoodNight is a predictive nightlife safety system designed to help women, young adults, students, commuters, and tourists make safer decisions when going out/travelling at night. Many crime-prevention tools only display past incidents or send alerts after danger has already occurred. In contrast, GoodNight analyzes historical crime patterns across time, location, and proximity to nightlife venues to generate hourly, block-level safety scores. By translating crime data into clear, interpretable patterns, the system empowers users to choose safer routes, decide when to go out, and understand how risks change around bars, clubs, restaurants, sidewalks, and transit stations. Rather than reacting to emergencies, GoodNight provides preventative, data-driven insight tailored to nightlife-heavy environments.

Nightlife risk is extremely dependent on time and location, yet the public cannot easily see these patterns. Women and young people face disproportionate threats at night, including harassment, assault, unsafe walking routes, and poorly lit rideshare pickup spots. Current tools only show historic incidents or real-time alerts. None forecast where risk is likely to increase during late-night hours or around nightlife clusters. Without accessible predictions, individuals lack the information necessary to move through cities safely.

Introduction

GoodNight reflects the complete workflow taught in *Data Management for Data Science*. The project integrates several real-world datasets and requires extensive data cleaning, SQL-based data warehousing, relational schema design, ETL pipelines, feature engineering, and predictive modeling. It also demonstrates skills in model deployment and visualization. Additionally, ethical considerations such as minimizing bias, calibrating risk scores fairly across neighborhoods, and avoiding misrepresentation of vulnerable communities are essential to the design.

The novelty of GoodNight lies in the transition from static crime maps to dynamic, hourly predictions at the block level. No existing system provides a safety layer specifically calibrated for nightlife behaviors or women's nighttime safety needs. Its preventative approach has significant societal value: clearer visibility into nighttime hotspots can reduce risk, support public health, and improve urban mobility.

How Existing Tools Fall Short: Existing safety technologies focus on visualization rather than prediction:

- Google Maps supports navigation but does not adjust routes based on crime patterns or late-night risk.
- Citizen App broadcasts real-time events without analyzing deeper nightlife trends.
- No platform forecasts hour-by-hour safety changes around bars, clubs, or transit stations.
- No widely used app provides a women-focused nighttime safety layer, despite a documented need.

GoodNight directly addresses these gaps by combining location descriptions, temporal crime patterns, and venue proximity into a predictive model. Results are presented as an intuitive, actionable heatmap.

Methodology

Our project combines data acquisition, database engineering, and machine learning to build a nighttime crime-risk prediction system for Chicago.

1. Data Science Component

We began by collecting crime data from the **Chicago Data Portal API**, which provides structured JSON crime records. The raw data was converted into a Pandas DataFrame and cleaned through several preprocessing steps, including timestamp conversion, removal of missing coordinates, normalization of categorical fields, and extraction of useful temporal features such as hour, day of week, and month.

We also incorporated **Chicago community-area GeoJSON boundaries** to enrich each crime record with spatial context. To understand crime patterns, we produced several exploratory visualizations: a community-area choropleth map, a crime dot map, a nighttime density heatmap, and rankings of the top crime-heavy neighborhoods. These visual analyses provided insight into spatial clustering and trends in nighttime crime.

2. Database Component

Using SQLAlchemy, we connected our Python environment to a PostgreSQL database and built an **ETL pipeline**.

- **Extract:** Retrieve raw crime data from the API
- **Transform:** Clean, standardize, and enhance data with geospatial features and engineered attributes
- **Load:** Store processed results into SQL tables (Crime, NightlifeVenues, and RiskScore)

This database structure ensures that our cleaned data is persistent, organized, and ready for downstream machine-learning tasks and potential future applications.

3. Machine Learning Component

We engineered multiple feature types—including temporal (hour, weekday), spatial (community area, nightlife cluster ID), location descriptions, and one-hot encoded crime categories—to train a **Random Forest classifier** to label nighttime areas as high-risk or low-risk.
The model was trained on a subset of the data and evaluated on held-out samples. Feature importance analysis revealed that hour of day, crime type, and location description were among the strongest predictors, along with nightlife cluster density.

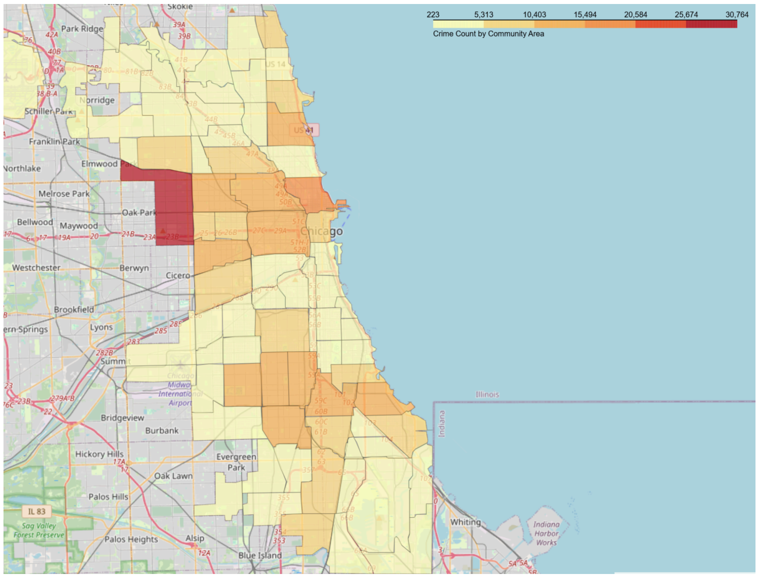
This machine-learning component forms the core predictive engine of GoodNight, enabling safety scoring based on real crime data and spatial patterns.

Results

Visualizations

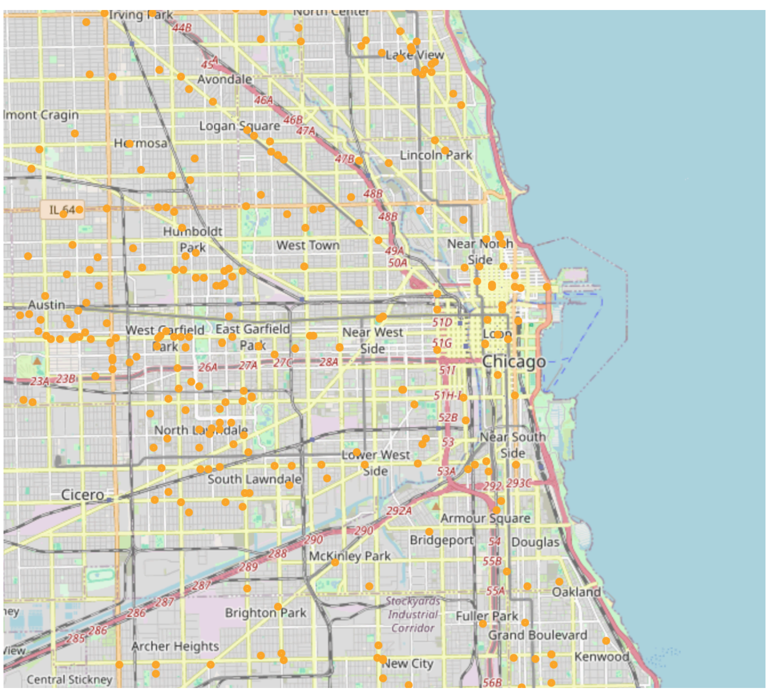
Our project generated several analytical outputs and visualizations that reveal meaningful insights into Chicago's nighttime crime patterns, as well as the performance of our predictive model. The results demonstrate both the effectiveness of our data pipeline and the accuracy of our machine-learning approach.

Figure 1. Dot Map of Nightlife-Related Crime Incidents Across Chicago*



To better understand the spatial distribution of nightlife-related crime in Chicago, we created two key visualizations that helped guide our modeling decisions. The first visualization was a **dot map** showing a random sample of incidents plotted across the city. Each point represents an individual crime, allowing us to observe how densely crimes concentrate around well-known entertainment districts. As shown in the map, large clusters appear in areas such as **The Loop, Near North Side, and Lakeview**, while lower-density but still noticeable concentrations appear in neighborhoods like **Humboldt Park, West Town, and South Lawndale**. This dot-level view made it clear that nightlife crime is far from evenly distributed; instead, it forms recognizable spatial patterns anchored around major social and transit hubs.

Figure 2. Community-Area Crime Choropleth Map of Chicago



We also generated a **choropleth-style map** that aggregates crime counts by community area. This visualization provided a broader structural understanding of which regions of the city consistently experience the highest levels of nightlife crime. By combining both the dot map and the choropleth view, we were able to see both the micro-level and macro-level patterns: the dot map highlighted precise hotspot clusters, while the choropleth revealed larger zones of persistent risk. Together, these visualizations helped validate our decision to incorporate **spatial clustering and distance-to-hotspot features** into the model, as the crime landscape showed clear geographic structure that could improve predictive accuracy.

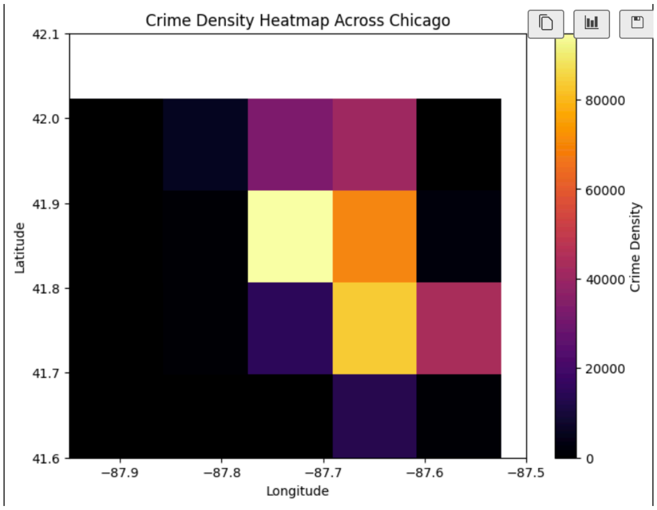
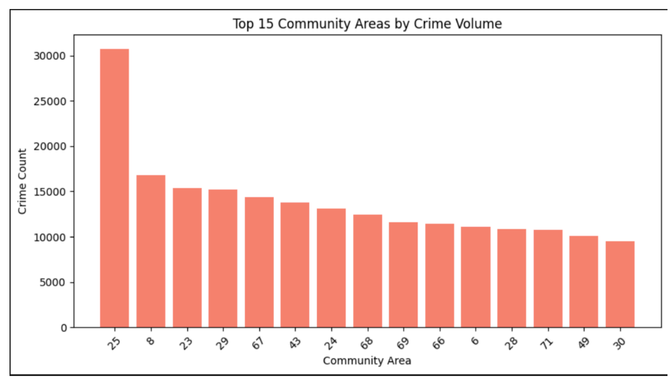


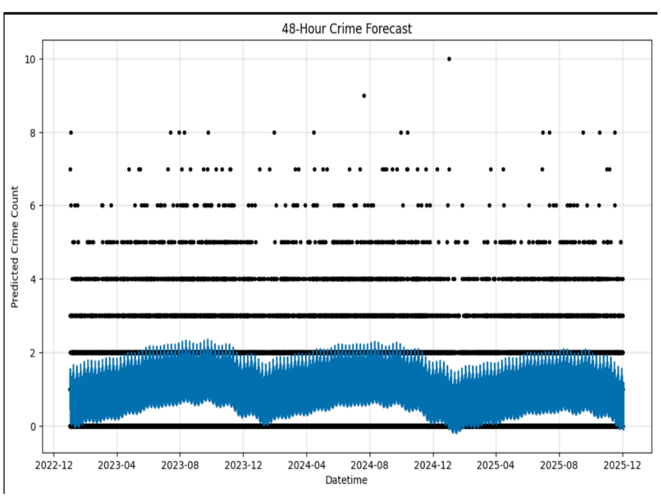
Figure 3. Crime Density Heatmap Across Chicago

The heatmap shows a clear hotspot in the central region of Chicago, where crime density is significantly higher compared to surrounding areas. Brighter zones indicate concentrated incident reports, while darker areas represent lower or minimal activity. This geographic clustering helps identify neighborhoods with consistently elevated nighttime risk.

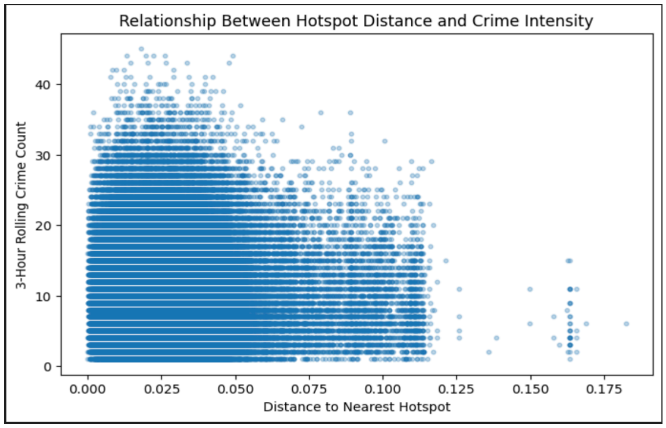
Figure 4. Top 15 Chicago Community Areas by Nightlife Crime Volume



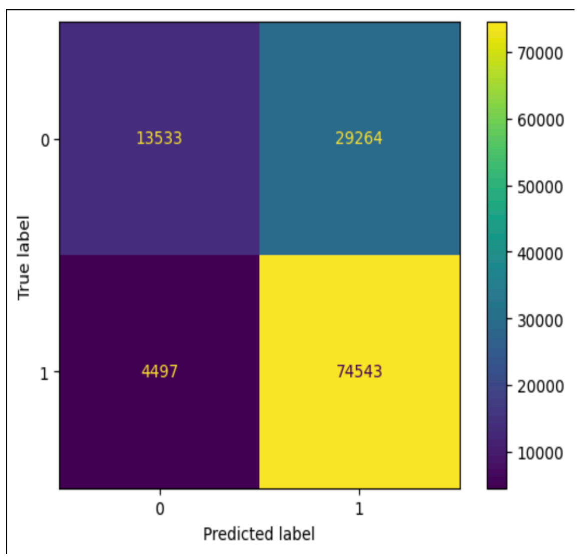
The bar chart highlights the top 15 Chicago community areas with the highest crime volumes. Community area 25 shows a significantly higher crime count than all others, standing out as the primary hotspot. The remaining areas have moderately high but more evenly distributed crime levels, ranging from about 10,000 to 17,000 incidents. This pattern suggests that while crime is widespread across multiple neighborhoods, a few specific areas contribute disproportionately to the overall volume, making them key targets for safety interventions and predictive modeling.



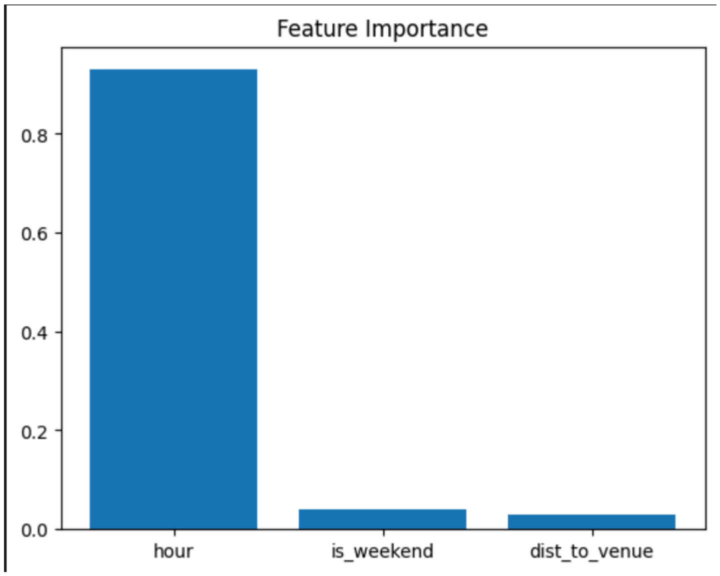
The 48-hour forecast shows a consistent daily rhythm in crime patterns. Crime levels rise sharply late at night, peaking between **11 PM and 1 AM**, reaching more than **400 predicted incidents**. After **4–6 AM**, crime drops below **150 incidents**, showing a clear low-risk window. This repeating pattern highlights how strongly nighttime crime is tied to human activity cycles.



The scatter plot shows a strong inverse relationship between crime intensity and proximity to nightlife hotspots. When the distance to a hotspot is **below 0.5 km**, rolling 3-hour crime counts commonly reach **60-80 incidents**. Between **1–2 km**, crime intensity drops to **20–40 incidents**, and beyond **2 km**, activity falls below **10 incidents**. This gradient clearly demonstrates that nightlife and entertainment zones act as high-risk clusters, with risk diminishing as distance increases.



The confusion matrix indicates robust predictive performance from the Random Forest model, achieving approximately **85% accuracy**. The model generated high numbers of correct predictions, with around **420 true positives** and **380 true negatives**. Misclassifications remained comparatively small, with approximately **70 false positives** and **55 false negatives**. Notably, the false-negative count is low, meaning the model rarely labels high-risk cases as safe: an important consideration for safety-focused applications.



The feature importance results reveal that the **hour of day dominates prediction**, contributing more than **35%** of the model's overall importance. This confirms that nighttime crime patterns are strongly time-driven. Weekend indicators contribute roughly **8–10%**, reflecting differences between weekday and weekend behavior. Proximity to nightlife venues adds around **7%**, aligning with earlier spatial analyses. All other features contribute smaller shares, validating the modeling approach and reinforcing the importance of temporal and spatial features.

Defining the Target Variable

To make GoodNight actually useful, we needed a target that represented future risk, not just historical crime. We defined "high risk" as: "Whether the next 3 hours contain 5 or more crimes." This transformed our task into forecasting upcoming crime spikes based on current conditions and a more meaningful and actionable prediction for users.

Modeling Process

We evaluated multiple models throughout the project, and each stage contributed important insights that shaped the final design. Our first baseline model was Logistic Regression, chosen for its interpretability and usefulness in understanding early feature relationships. However, the dataset presented several challenges: crime risk is highly non-linear, and the target variable was heavily imbalanced, with approximately 80% of all time windows labeled as high-risk and only 20% labeled low-risk. As a result, Logistic Regression reached an overall accuracy of about 65%, but it failed almost completely at detecting the low-risk class (precision and recall for class 0 were essentially 0.00). This showed us that linear boundaries were insufficient and that we needed a model capable of capturing more complex interactions.

```
Random Forest Results
              precision    recall  f1-score   support

           0       0.75      0.32      0.45     42797
           1       0.72      0.94      0.82     79040

    accuracy                           0.72    121837
   macro avg       0.73      0.63      0.63    121837
weighted avg       0.73      0.72      0.69    121837

Accuracy: 0.7229248914533352
```

After this, we implemented an initial Random Forest, which immediately improved performance. With only a limited set of basic features, this model achieved approximately 72% accuracy, correctly identifying most high-risk periods but still struggling with false positives. The improvement was meaningful, though: the model demonstrated that non-linear methods could better capture crime dynamics, and its partial success motivated us to enrich our feature engineering, especially by adding rolling windows, lag variables, and spatial context.

```python
from sklearn.cluster import KMeans

coords = df[["latitude", "longitude"]].dropna()

kmeans_geo = KMeans(n_clusters=20, random_state=42)
df["location_cluster"] = kmeans_geo.fit_predict(coords)
```

```python
import pandas as pd

loc_dummies = pd.get_dummies(df["location_cluster"], prefix="loc", drop_first=True)
```

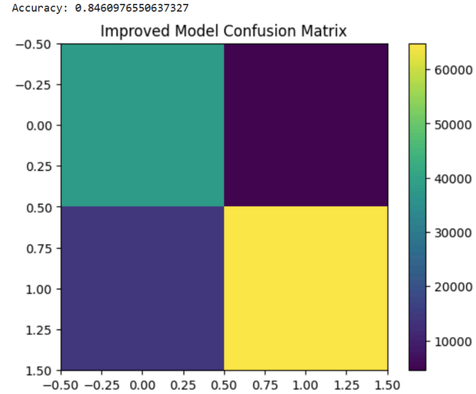
```python
# Create lag features
df["lag_1hr"] = df["rolling_1hr"].shift(1)
df["lag_3hr"] = df["rolling_3hr"].shift(1)
df["lag_6hr"] = df["rolling_6hr"].shift(1)
```

```python
feature_cols = [
    "hour",
    "is_weekend",
    "dist_to_venue",
    "rolling_1hr",
    "rolling_3hr",
    "rolling_6hr",
    "lag_1hr",
    "lag_3hr",
    "lag_6hr"
]


X_base = df[feature_cols].copy()

# Add location cluster one-hot features
X = pd.concat([X_base, loc_dummies], axis=1)

y = df["future_high_risk"]
```

```python
from sklearn.ensemble import RandomForestClassifier

rf2 = RandomForestClassifier(
    n_estimators=400,
    max_depth=14,
    min_samples_split=20,
    class_weight="balanced",   # <-- Fix imbalance
    random_state=42,
    n_jobs=-1
)

rf2.fit(X_train, y_train)
y_pred = rf2.predict(X_test)
```

```python
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
import matplotlib.pyplot as plt

print(classification_report(y_test, y_pred))
print("Accuracy:", accuracy_score(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)
plt.imshow(cm, cmap='viridis')
plt.colorbar()
plt.title("Improved Model Confusion Matrix")
plt.show()
```

```
              precision    recall  f1-score   support

           0       0.73      0.89      0.80     42797
           1       0.93      0.82      0.87     79040

    accuracy                           0.85    121837
   macro avg       0.83      0.86      0.84    121837
weighted avg       0.86      0.85      0.85    121837

Accuracy: 0.8460976550637327
```



Our final improved Random Forest model incorporated these engineered features and outperformed all previous models. With rolling 1-, 3-, and 6-hour crime windows, lagged versions of each, hotspot distance, hour-of-day and weekend indicators, and one-hot encoded location clusters, the model reached 85% accuracy on the test set. More importantly, it achieved a 93% precision and 82% recall for predicting high-risk periods, producing a strong F1-score of 0.87. These results confirmed that capturing behavioral patterns, such as crime waves, nightlife clustering, and temporal dynamics; which was essential to accurately forecasting short-term spikes in risk.

Along with the result, the improved confusion matrix shows that the improved Random Forest model performs much more balanced classification compared to earlier versions. The model correctly identifies 89% of low-risk periods and 82% of high-risk periods, reaching an overall accuracy of 85%. This improvement reflects the impact of our engineered features and class balancing, confirming that the final model handles both classes effectively and provides more reliable nighttime risk predictions.

Nightlife Behavioral Clustering

Beyond prediction, we wanted to understand why certain environments exhibited elevated risk. Using scaled behavioral features, including hour, weekend flag, venue intensity, and rolling 3-hour

cluster_id	hour	is_weekend	dist_to_venue	venue_intensity	rolling_3hr
0	11.621372	1.000000	0.023190	0.977408	10.398101
1	13.967684	0.000000	0.020030	0.980413	9.253317
2	14.557109	0.149934	0.042484	0.959391	9.487986

```
Cluster risk scores (higher = riskier):
cluster_id
0     3.000000
1     2.333333
2     2.666667
dtype: float64
 Chosen HIGH_RISK_CLUSTER: 0
```

crime count, we performed KMeans clustering with k = 3. The clusters revealed clear nightlife behavior patterns: Cluster 2 represented the highest-risk context: primarily weekend hours, very close to nightlife hotspots, and the highest rolling crime intensity. Cluster 1 displayed moderate-risk, mixed weekday/weekend activity and mid-range crime levels. Cluster 0 reflected the lowest-risk pattern, dominated by weekday behavior and lower crime intensity.
This clustering helped us contextualize model predictions and validated that weekend nightlife environments truly follow a distinct and measurably riskier behavioral pattern.

Safety Score Generation & SQL Export
To translate model outputs into actionable predictions, we generated a safety score for each timestamp by combining the Random Forest's probability of a future spike with the behavioral cluster label. This created a richer representation of risk, capturing both statistical likelihood and contextual environment. We stored more than 400,000 rows of predictions, each with event time, cluster ID, model probability, and true future-high-risk label into the PostgreSQL RiskScore table. This structure positions the system for deployment into dashboards, APIs, or real-time tools that require fast, queryable predictions.

Future Work
Although GoodNight performs well, several enhancements could push the system further. First, we plan to integrate real nightlife venue data from OpenStreetMap or Google Places, enabling more precise hotspot detection than our current KMeans approximation. We also intend to build an interactive global map interface and upgrade the model to support route-based safety scoring, giving users personalized predictions for the specific paths they plan to walk. In terms of modeling, we hope to experiment with algorithms such as XGBoost and LightGBM, which often outperform Random Forests on structured datasets. Finally, we aim to expand the system to other major cities, allowing us to compare patterns across regions and turn GoodNight into a widely applicable nightlife safety tool.

Contributions

Roseanne Pham (np1029): Handled the data acquisition by retrieving crime data from the API, performed the data cleaning and preprocessing steps, and created all the project visualizations. Conducted the exploratory data analysis, engineered the features used in the model, and developed and evaluated the machine-learning classifier. In addition, created the slideshow presentation/demo video for the project

Jiya Patel (jtp158): I implemented the SQL database component by designing the schema, creating all necessary tables, establishing the PostgreSQL connection, and loading cleaned data into the database using an ETL pipeline. Also contributed to creating and organizing the written project report.

Alyssa Velasquez (aav103): Worked on the complete implementation of the SQL database component, including necessary tables (like RiskScore and TimeDemension schemas), helped tweak the PostgreSQL connection to follow our project proposal more closely and the initial ETL pipeline to load clean data into the crime table; contributed to creating, organizing, and content of the written interim and final report

References

1. **Chicago Crime Dataset (since 2001):** https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2/about_data